

MA1513 Weekly Self Practice Set 4

1. (a) Given that $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a linear transformation such that

$$T \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \quad T \left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}, \quad T \left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}.$$

Write down the standard matrix $\mathbf{A}$ of T . Briefly explain how you find $\mathbf{A}$.

- (b) Without solving characteristic polynomial or using MATLAB, find all the eigenvalues of the standard matrix $\mathbf{A}$ in part (a).
- (c) Find a basis for each of the eigenspaces of $\mathbf{A}$ associated with all the eigenvalues (without using MATLAB). Show your working.
- (d) Using MATLAB with format short option for this part.
- (i) Type the command `>> [P1 D1]=eig(A)`. (You will get a matrix $P1$ where the columns are eigenvectors; and a diagonal matrix $D1$ where the diagonal entries are eigenvalues.) Write down the two columns of $P1$ that give the eigenvectors corresponding to the larger eigenvalue of $\mathbf{A}$.
- (ii) Type the command `>> [P2 D2]=eig(sym(A))`. Write down the column of $P2$ that gives the eigenvector corresponding to the larger eigenvalue of $\mathbf{A}$.
- (iii) Explain the differences that you have observed in (i) and (ii).